Ralphie’s Innovations

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1. INTRODUCTION

Following the preliminary design review (PDR) submission, the product design shifted from the initial goal of creating a portable charger to prioritizing charging within a classroom setting. As a result of this shift, prototyping has primarily focused on the product's use in stationary settings due to its necessary size for functionality and safety.

Ralphie's Innovations, an intro design course team, is developing a product called the SolarBoost, a solar-powered charger to charge your phone at any time the need arises, in an environmentally friendly manner. The product is intended for college students who heavily rely on their devices for navigation, communication, and school work in their day-to-day lives.

SolarBoost was designed with environmental sustainability in mind when addressing the common student need to frequently charge their cellular devices. There are already products on the market that produce charge through solar power, however, these products are non-aesthetic and expensive. Additionally, phone-charging products that utilize charging stations and compact setups exist in the market. However, these utilize standard charging methods that are not environmentally friendly and sustainable. SolarBoost aims to bridge the gap and establish a charger that improves on both aspects simultaneously.

This report will include an overview of the design criteria, prototyping, and the final design and CAD models. Within the design criteria, there will be an analysis of what was deemed more important than others, including the different components of functionality, like charging speed, as well as other factors such as size and weight. A list of requirements will be placed showing what SolarBoost needs to accomplish to be considered successful.

Three designs were brought to the prototyping phase, a model with a stand, one with a clip, and one with a suction cup. Details for those, as well as CAD models are provided in this section. The next section, Prototyping, will detail the procedures taken to build the SolarBoost charger. The step-by-step process will be spoken about in detail here to demonstrate the effort put into designing SolarBoost, as well as any problems encountered and overcame.

The final section, Final Design and CAD Models, will then detail the final product and how it was manufactured. Subjects such as how all the components were put together to form a functioning product, as well as a layout of the circuit and everything else inside the CAD shell will be discussed. Overall, the CDR's goal is to provide an overview of the creation of the SolarBoost charger from start to finish.

2. DESIGN CRITERIA

This section focuses on the design criteria for the project, including a detailed discussion of the measurable criteria and an evaluation of the trade space analysis. Three initial candidate designs are presented, highlighting the pros and cons of each. This use of analysis demonstrates a strategic way of choosing a final design that aligns with the project goals.

2.1 TRADE SPACE ANALYSIS

Before beginning to develop initial prototypes, consideration of several criteria was necessary to drive the design. Regarding the product's physicality, the safety, weight, and size were considered. Battery charging speed and phone charging speed assisted in describing the desired efficiency of the product. Aesthetic and cost were qualities that might impact user attraction. Below in Table 1, is the complete Weighted Design Criteria table.

	Cost	Safety	Weight	Size	Battery charging speed	Phone Charging Speed	Aesthetic	Row Total	Normalized Value
Cost		0	0	0	0	0	2	2	.020
Safety	5		5	5	5	5	5	30	.286
Weight	5	0		1	4	2	3	15	.143
Size	5	0	4		4	3	4	20	.190
Battery charging speed	5	0	1	1		0	5	12	.114
Phone charging speed	5	0	3	2	5		5	20	.190
Aesthetic	3	0	2	1	0	0		6	.057

Table 1: Weighted Design Criteria

The design must prioritize some key requirements to ensure user safety and functionality. It should provide charging speeds relatively close to a regular charger while addressing any overheating issues. Additionally, the design must be able to safely produce 5 volts without causing damage to the user or the phone. Finally, a lightweight design must be maintained; the team aimed for under one pound. The combination of these requirements ensures a safe and efficient solution.

Relative to the other design criteria, cost and aesthetic scored the lowest meaning the design team felt that the consumer price and visual appeal of the product were least important compared to the other criteria. While the team must still consider these features, the safety, speed, and size of the product were rated most important. At times, less aesthetic or more expensive decisions needed to be made to increase the functionality of different criteria.

Ultimately, the final product ended up larger than the team originally planned for because after considering the safety and efficiency, the sizing criteria had to be reevaluated.

2.2 INITIAL CANDIDATE DESIGNS

Design 1: Solar Panel Holder

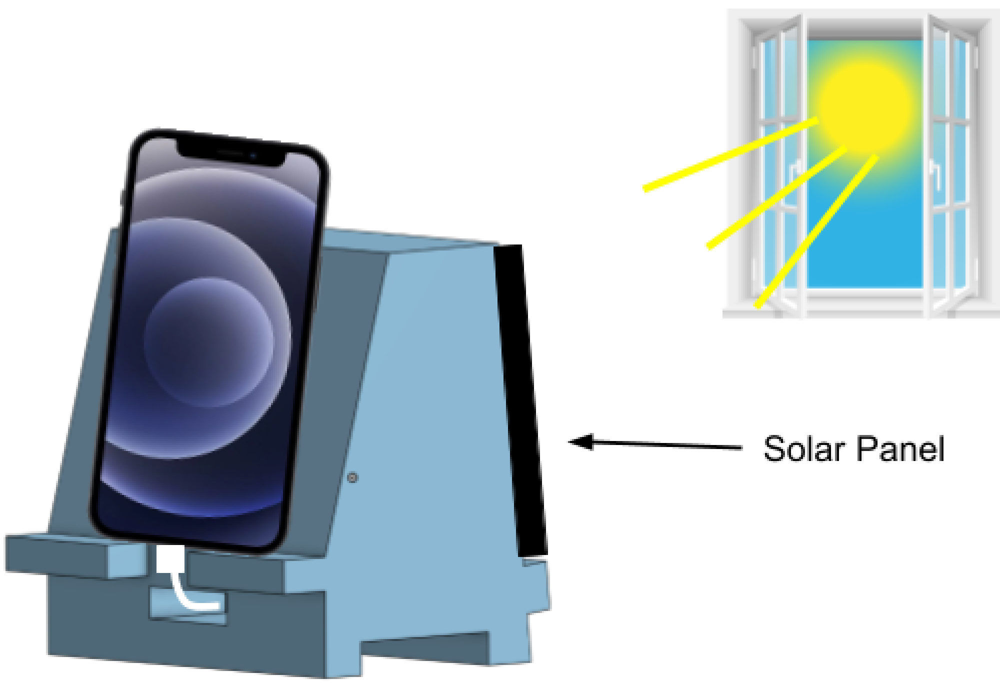

Figure 1: Solar Panel Holder

The Solar Panel Holder, shown in Figure 1, is the most compact and simple out of all designs considered. Its structure is very simple: the body houses the battery and circuit, and the ledges on either side hold the phone and solar panel. The design features no wiring outside of the structures allowing for complete safety of the circuit inside. There is a great amount of concern with this design associated with the solar panel capturing the maximum amount of sunlight because of the fixed position the solar panel is in. Lack of sunlight ultimately affects the phone and battery charging speed.

Design 2: Solar Panel Clip

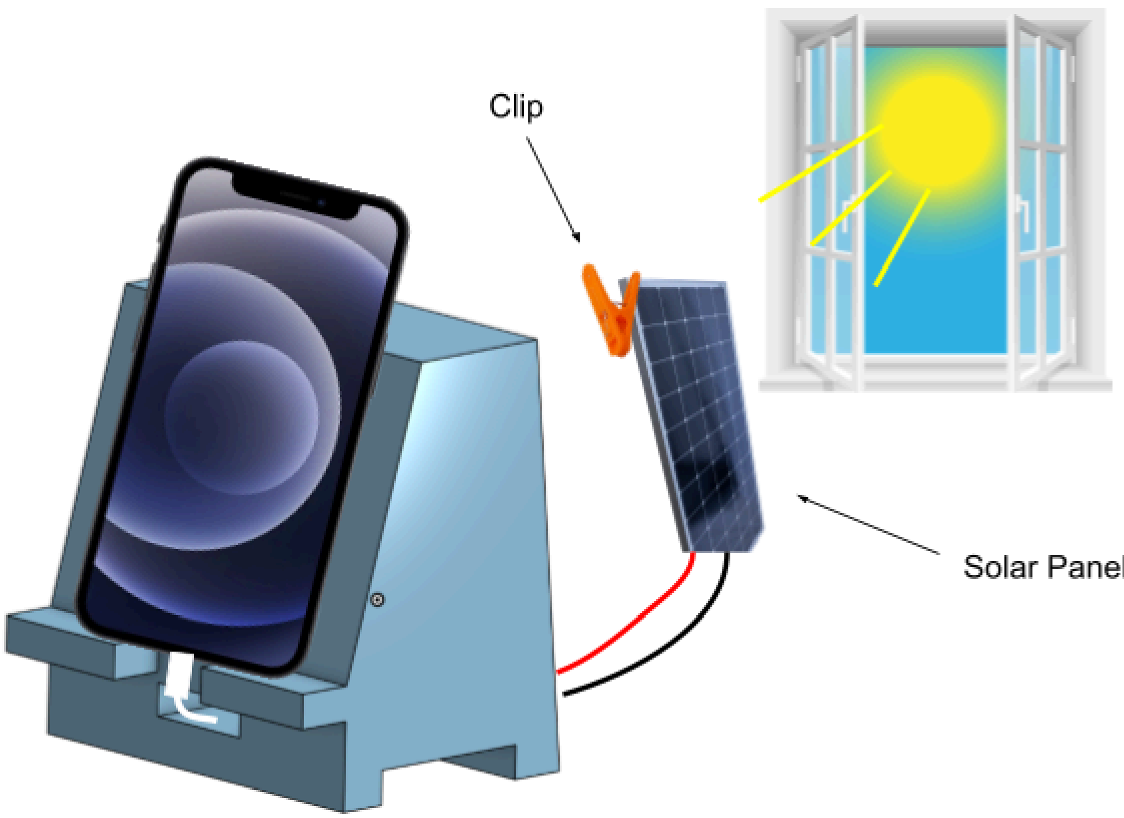

Figure 2: Solar Panel Clip

The design of the Solar Panel Clip is similar to the above design, with one key difference: the solar panel can be detached from the holder. This added feature allows users to reposition the panel for optimal exposure to direct sunlight. This offers greater flexibility, as shown in Figure 2. The Solar Panel Clip is a versatile design that can be clipped onto a user's backpack while walking to class or the back of a computer as they are working. Nevertheless, with its wide range of applications, there are some potential concerns, such as the wire getting snagged or the clip scratching the back of the user's computer, especially given its mobile nature.

Design 3: Suction Cup on Solar Panels

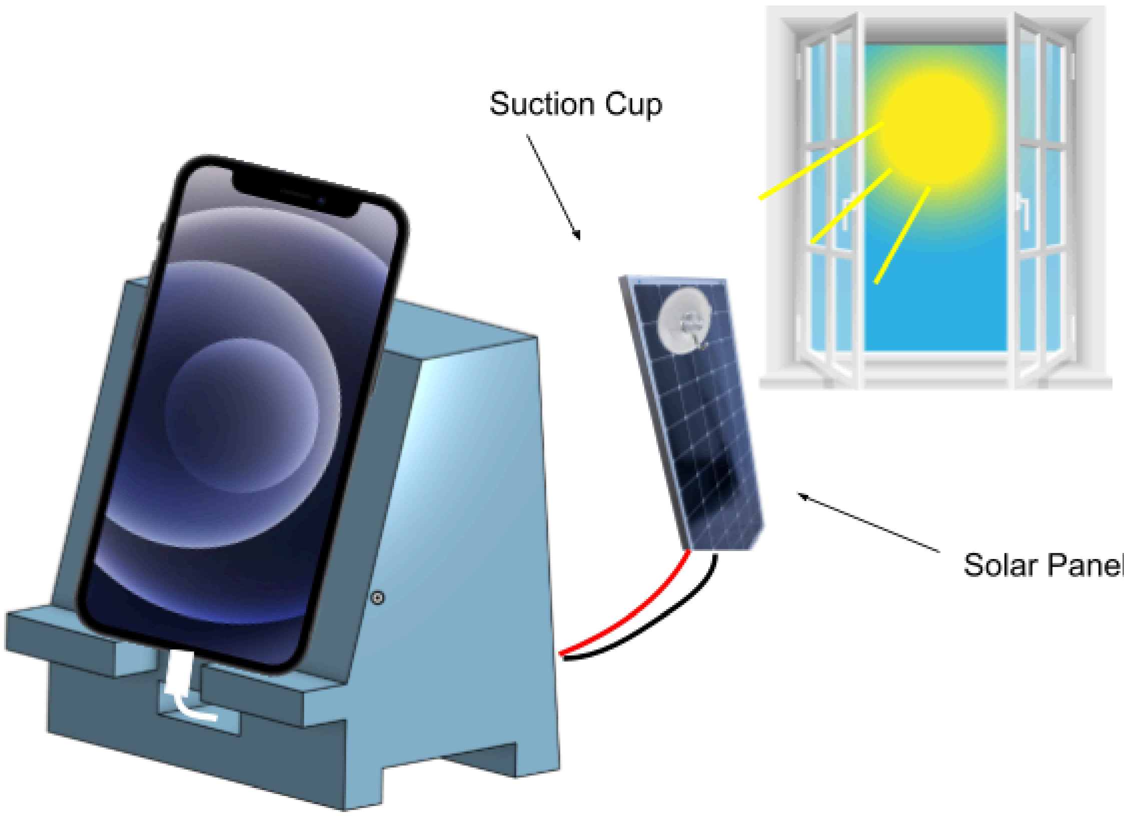

Figure 3: Suction Cup on Solar Panel Phone Holder

The Suction Cup on Solar Panel Phone Holder design allows for the best battery charging efficiency. Because the solar panel is intended to be stuck to a window as depicted in Figure 3, it will have access to the most direct sunlight in comparison to the other designs, resulting in faster battery charging. Although similar to the clip-on model, this design is intended to be more stationary – it must be placed and left near a windowsill – making its exposed wires less of a danger.

2.3 WEIGHTED DESIGN MATRIX

Criteria	Normal Priority Value	Design Alternative #1 (Solar Panel Holder)	Design Alternative #2 (Solar Panel Clip)	Design Alternative #3 (Suction Cups Solar Panels on Window)
Cost	0.020	3 | 0.060	2 | 0.040	2 | 0.040
Safety	0.286	5 | 1.43	3 | 0.858	5 | 1.43
Weight	0.143	4 | 0.572	3 | 0.429	4 | 0.572
Size	0.190	3 | 0.570	3 | 0.570	3 | 0.570
Battery Charging Speed	0.114	2 | 0.228	2 | 0.228	2 | 0.228
Phone Charging Speed	0.190	4 | 0.760	4 | 0.760	4 | 0.760
Aesthetic	0.057	4 | 0.228	3 | 0.171	4 | 0.228
Total		3.85	3.06	3.83

Table 2: Rating of Criteria | Normalized Criteria Value

As seen above in Table 2, the solar panel holder numerically fits all of the criteria the best, but a close second was the suction cup solar panel. The team decided upon the suction cup design due to its uniqueness from what is on the product market already and the challenge it would provide. The suction cup solar panel still reaches the criteria for the desired amount to ensure the team's goals and desires are met.

3. PROTOTYPING

This section identifies potential problems early on in the development process to save time, resources, and money, as well as refine the design as needed. It was important to ensure improvements were being made before moving forward with the final product. As expected, many issues presented themselves at the beginning of the prototyping process. Issues with the lightning phone charger and battery charging module were major areas of concern for the team.

3.1 CIRCUIT

In an Apple lightning phone charger, there are four wires, a positive pink wire, a negative gray ground wire, a green negative data wire, and a white positive data wire – see Appendix A (Figure A-1). When it comes to charging a phone, the pink and gray wires are responsible for sending power to charge the phone and the data wires don't need to be used unless data needs to be transferred. The prototype started as a power-only charger; this approach quickly became a problem because the phone would not trust the charger and as a result, the phone would not charge. After some further research, the data wires would need to be connected in some fashion to the circuit for the phone to trust the charger. The data connection is designed to verify the authenticity and compatibility of the charger before allowing it to charge. Data wires check specific parameters, such as voltage and current levels, and they communicate to the phone that the charger is safe and will not cause damage to the phone.

On many different occasions, the battery charger module was observed to be hot to the touch, a concern that was expected to be run into. As long as the module was not hot to the touch and could not burn a person, then it was alright to be left alone. Out of precaution, a heat sink was placed underneath the module. The heat sink allows for the heat to be absorbed and then dissipated in a way that regulates the temperature of the charging module.

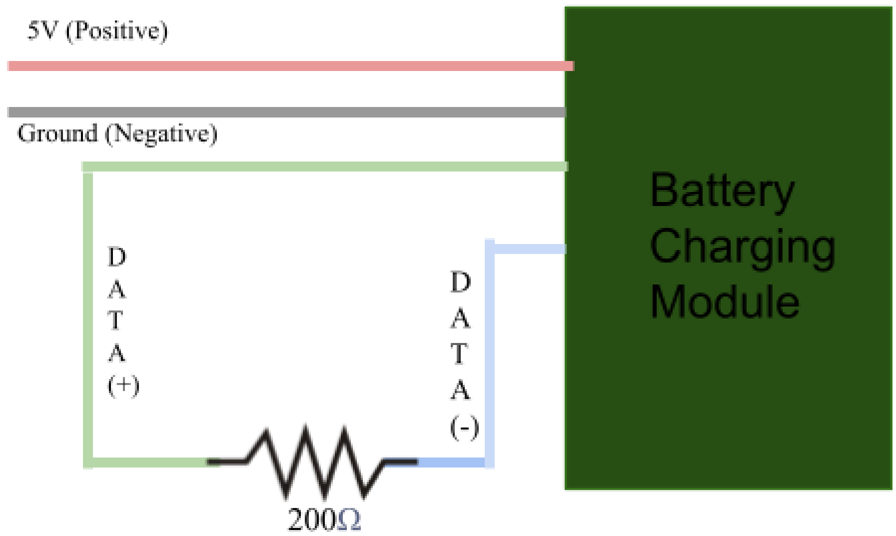

Figure 4: Circuit setup with the inclusion of the data wires

As shown above, in Figure 4, the team did extensive research to determine the circuit layout and functionality for an optimal phone charger.

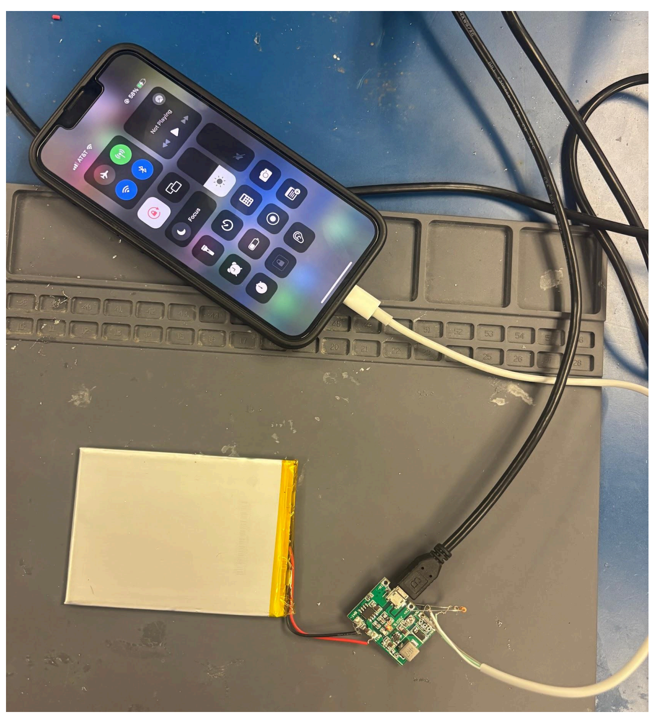

Figure 5: Testing the battery charging circuit on a phone

Figure 5 above demonstrates the battery and charging module’s ability to increase the charge of an iPhone.

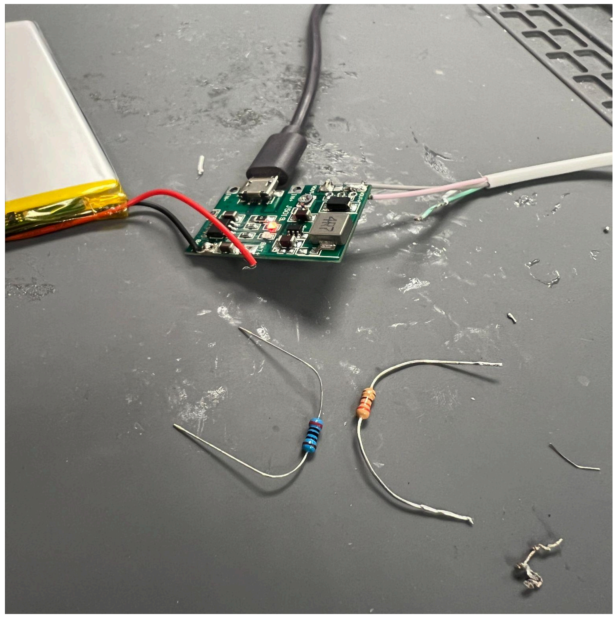

Figure 6: Testing different resistors

As shown in Figure 6, the team tested a 200-ohm resistor and a 220-ohm resistor. The team started with a 220 ohm resistor which enabled trust with the phone but failed to charge the phone anywhere close to the rate of a regular charger. Then, the team used a 200-ohm resistor which quickly connected with the tested iPhones and charged at a rate close to a regular phone charger.

Figure 7: Solar panel charging iPhone

Figure 7 demonstrates that the charging ability of the solar panel was tested to ensure the model was not merely taking charge from what was leftover in the rechargeable battery. The team wanted to determine whether the solar panel could be used to actively charge the phone, or if it needed to charge the battery a certain amount before the phone could even receive the charge. As seen in Figure 7, the phone received a charge while the solar panel was receiving sunlight.

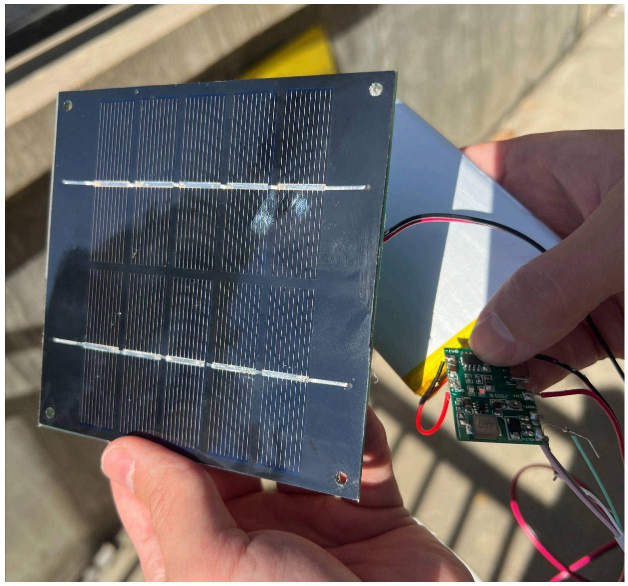

Figure 8: Solar Panel Charging

As seen in Figure 8, the red light shows that the solar panel is actively charging the rechargeable battery. The team later did a test to find the charging current of the solar panel by using a voltmeter, which was measured to be 0.8 A. The battery was specified to have a battery capacity of 2.5 Ah. It was calculated that the battery would take 3.125 hours to charge. This was found using the formula: charging time equals battery capacity divided by the charging current. This time could vary depending on how sunny it is outside and how long the panel gets direct sunlight.

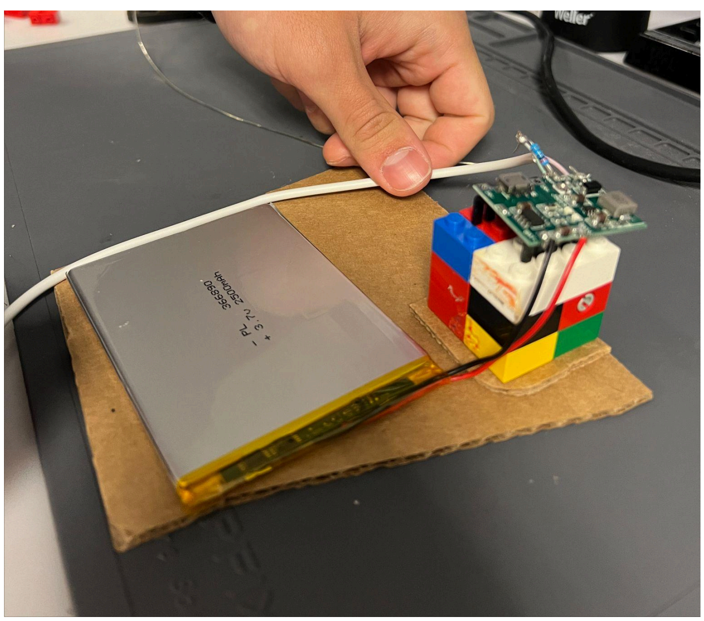

Figure 9: Inside set-up of battery charging circuit

Lastly, Figure 9 depicts how the circuit will sit inside the shelling. A Lego stand was created to support a heat sink which was used to dissipate thermal energy created by the charging module. This structure had to be elevated so it would not damage the casing.

3.2 SHELL

Figure 10: Front view of card-board holder prototype

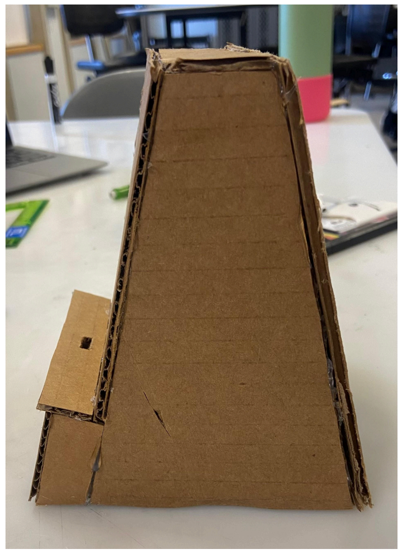

Figure 11: Side view of card-board holder prototype

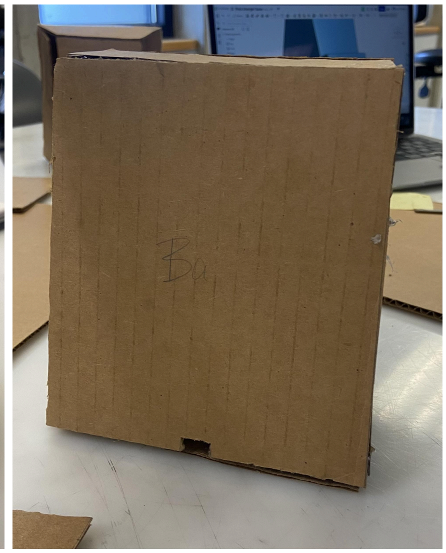

Figure 12: Back view of card-board holder prototype

After consulting the design criteria and decision matrix to establish which features must be included in the final design, the three major components of the structure were a phone stand, space to store the solar panel, and space inside to safely house the circuitry. As shown above, in Figures 10, 11, and 12, the team built a cardboard prototype to test how each portion functioned in the necessary ways. This way, it could be easily ascertained whether or not the current design would work without wasting expensive resources. The design was 3D printed because it included mostly non-right angles that would be difficult to physically reproduce using other methods. The PLA material used in 3D printing is also thin and lightweight, reducing the overall bulkiness of the product. It additionally gives the product a smooth and sleek look that may be difficult to achieve with other materials.

4. FINAL DESIGN AND CAD MODELS

This section will address the final designs of SolarBoost, splitting it into the electronics and the physical structure. Each will be thoroughly described giving ideas of its function and purpose in terms of achieving the design requirements.

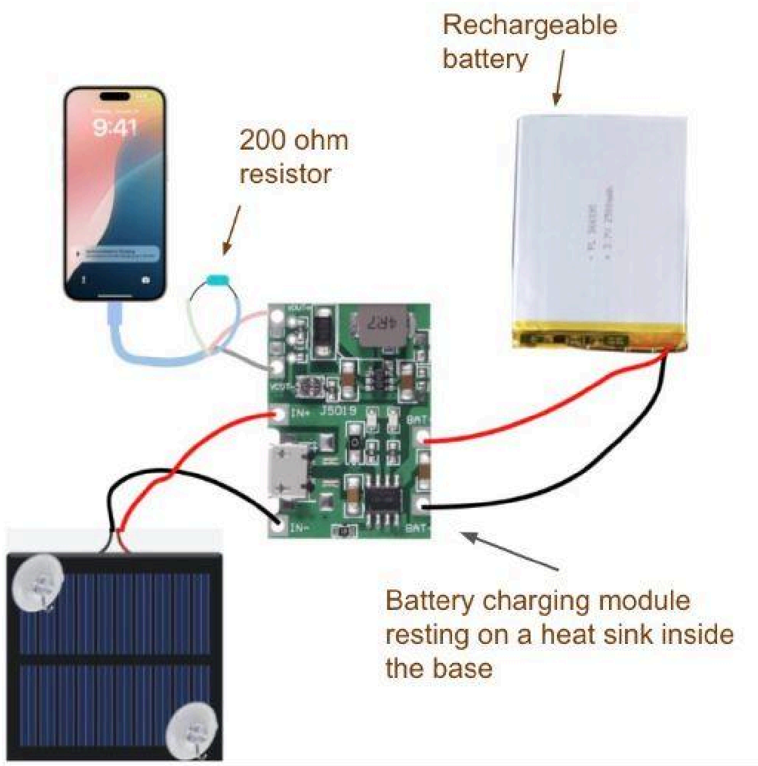

Figure 13: Final circuit design

The final circuit design integrates a solar panel, a battery charging module, a lithium rechargeable battery, a 200-ohm resistor, and a lightning cable charger to create an efficient and user-friendly charging system. The solar panel serves as the main power source, delivering a charging current of 0.8 A, which fully charges the lithium battery in approximately three hours. Once the battery is charged, the charging module regulates the output to exactly 5 volts, ensuring safe and efficient phone charging. This adjustment was achieved by using a voltmeter and turning a knob on the module to precisely restrict the voltage output. At this regulated voltage, the system charges phones at a speed comparable to standard chargers – see Appendix A (Chart A-1).

The design also incorporates a 200-ohm resistor to facilitate communication with the user's phone. This resistor enables the phone to recognize the charger as a trusted device, displaying the message "Accessory Connected" upon connection. This communication ensures that any user will be satisfied, as their device will connect and charge. Together, these components deliver a reliable, solar-powered charging solution optimized for convenience and efficiency.

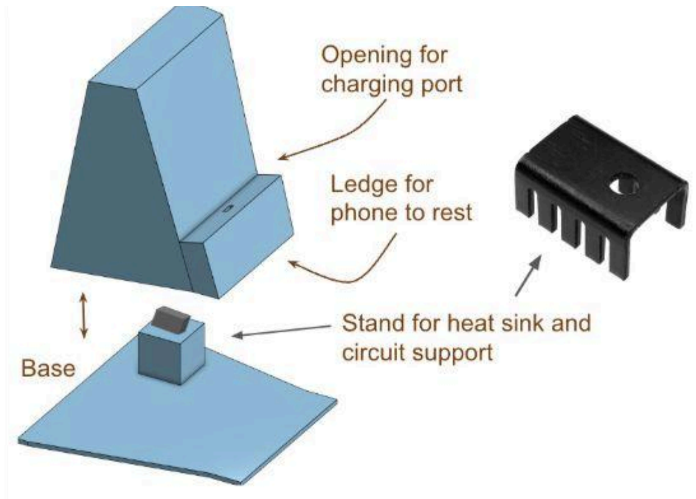

Figure 14: Structure Design

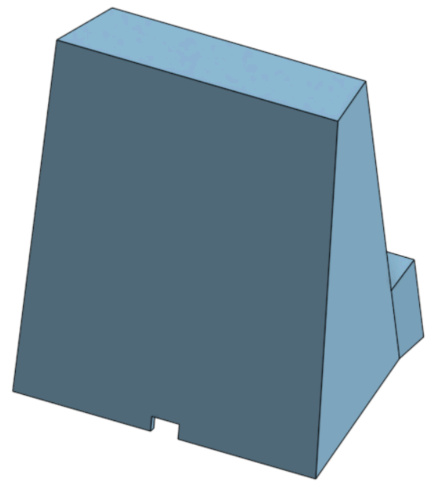

Figure 15: Back of Structure Design

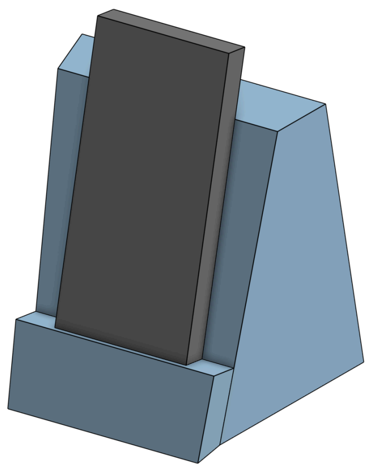

Figure 16: Model of product usage

The final structure design addresses concerns with overheating. As the phone is being charged, there is a noticeable increase in the temperature of the module. To address this, the module is placed on top of a heat sink to allow the heat to be dissipated. There is also a small stand for the entire setup, ensuring the heat does not transfer directly to the base of the structure.

The final design has a footprint of 5 by 5.5 inches, and a height of 6 inches. These dimensions were optimal for allowing the solar panel to be stored on the back of the holder. The circuit design described above will be compacted within this structure, with the charger looping through a tiny hole on the ledge and the solar panel's cord coming out of the back. This design ensures durability, as the circuit is well-protected within the structure. Additionally, the back of the charger is slanted, allowing the solar panel to rest easily on the back.

Ultimately, SolarBoost ensures that the user's phone will be safely and efficiently charged in an environmentally friendly way. It produces enough volts to charge the user's phone and addresses concerns about overheating to safely charge it. The product is also very unique and lightweight and ready to be implemented within any college campus.

5. APPENDIX A

Key:
Red: Pink
Black: Grey
Green: White
White: Green

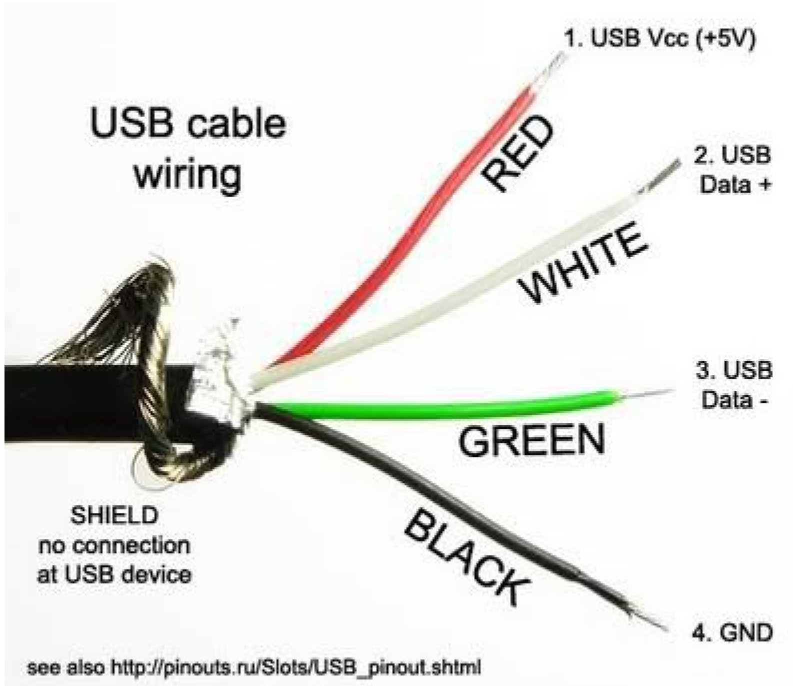

Figure A-1

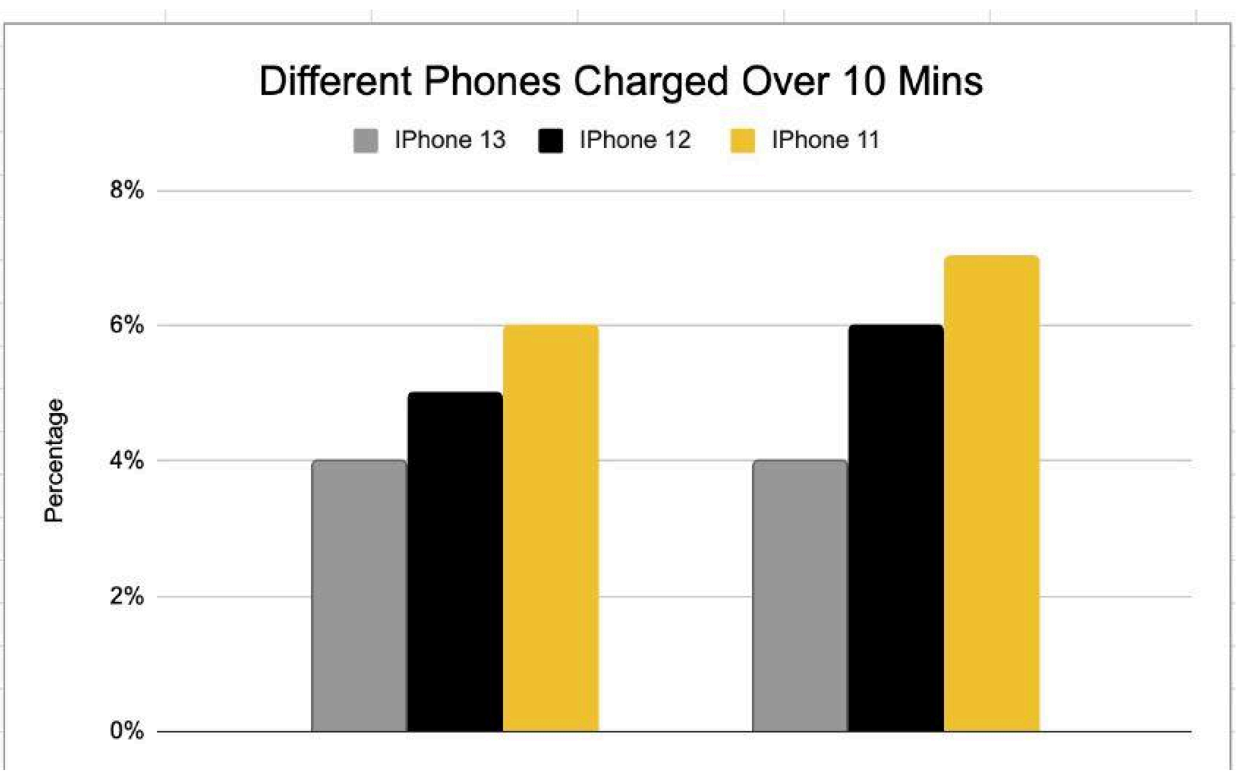

Chart A-1

6. REFERENCES

Ada, L. (2013, June 11). *iCharging | Minty Boost*. Adafruit Learning System. Retrieved November 13, 2024, from https://learn.adafruit.com/minty-boost/icharging